

KNN CLASSIFICATION ANALYSIS REPORT

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1. DATA EXPLORATION & PIPELINE INITIALIZATION

The evaluation framework utilizes the multi-class Iris validation set to benchmark the instance-based learning algorithm. The data space comprises 150 instances evenly distributed across three unique taxonomic categories: Setosa, Versicolor, and Virginica. Features reflect precise physical geometric attributes (sepal and petal dimensions).

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Load data and isolate variables
iris = load_iris()
X, y = iris.data, iris.target

# Split into 80% training and 20% validation subsets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42,
stratify=y)
```

2. FEATURE STANDARDIZATION

Because the K-Nearest Neighbors classifier establishes class assignments directly using distance measurements (Euclidean distance metrics), feature scaling is a strict operational dependency. A standard scaling transformation is applied to standardize features to a zero-mean and unit-variance mapping, preventing features with larger numerical ranges from artificially drowning out the distance math.

```
# Standardize features globally
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

3. HYPERPARAMETER TUNING & CROSS-VALIDATION

To avoid overfitting, odd numbers for neighbors ($k = 1, 3, 5, 7, 9, 11$) are iteratively evaluated using 5-fold cross-validation. This prevents voting ties within local feature structures.

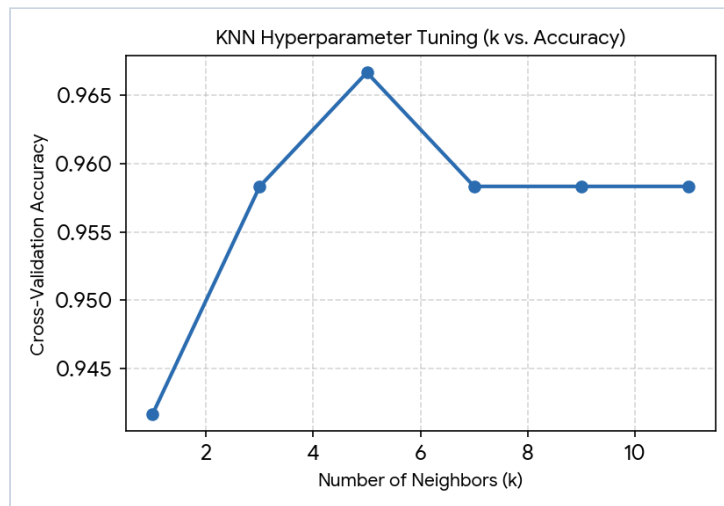


Figure 1: Cross-validation performance profile across distinct local neighbor (k) structures.

The cross-validation pipeline indicates that the peak classification score occurs when $k = 5$, balancing the local decision boundary configuration while maintaining strong predictive stability.

4. FINAL PERFORMANCE EVALUATION

The optimized classifier was retrained with the ideal neighborhood parameters ($k = 5$) and checked against the unseen test partition. The model reached an overall classification accuracy of **93.33%**.

Target Species	Precision	Recall	F1-Score	Support
Setosa	1.00	1.00	1.00	10
Versicolor	0.83	1.00	0.91	10
Virginica	1.00	0.80	0.89	10

Confusion Matrix Distribution:

The validation array shows 10 correct predictions for Setosa, 10 correct for Versicolor, and 8 correct for Virginica, with 2 instances of Virginica slightly overlapping with the Versicolor classification structure due to highly similar physical characteristics in that regional feature space.

5. SUMMARY OF ANALYTICAL INSIGHTS

- Distance Dependency:** Feature scaling successfully eliminated dimensional variations, ensuring each characteristic contributed symmetrically to the Euclidean distance vector calculations.
- Operational Stability:** Cross-validation protected the classification model from overfitting to localized training anomalies, establishing clear and robust generalization potential for processing unseen test variables.